

Bitcoin

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Acknowledgement

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Network Types

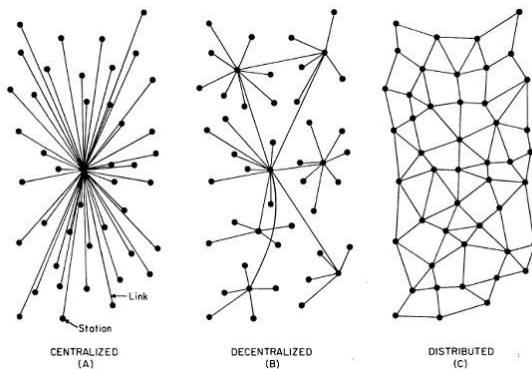


Figure: Types [From Mari Eagar]

Address Computation (Bitcoin)

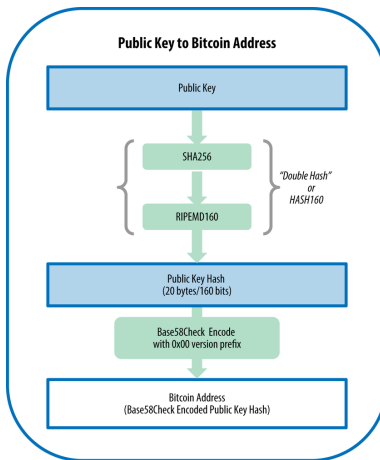
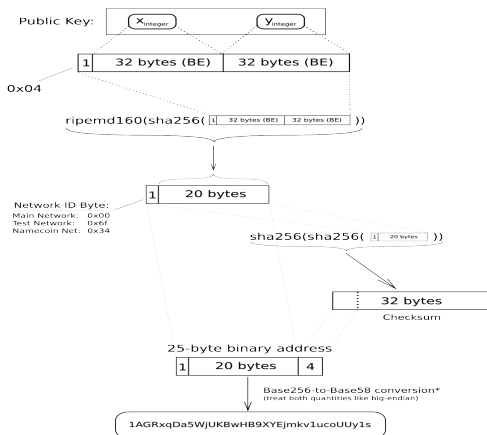


Figure: Copied from https://en.bitcoin.it/wiki/Technical_background_of_version_1_Bitcoin_addresses

Address Computation (Bitcoin)

Elliptic-Curve Public Key to BTC Address conversion



*In a standard base conversion, the 0x00 byte on the left would be irrelevant (like writing '052' instead of just '52'), but in the BTC network the left-most zero chars are carried through the conversion. So for every 0x00 byte on the left end of the binary address, we will attach one '1' character to the Base58 address. This is why main-network addresses all start with '1'.

etotheipi@gmail.com / 1Gffm7LKXcNFPrtxy6yF4jBoe5rVka4sn1

Node Type

- Light Node - Stores only header.
- Full Node - Store complete data and participate in mining.

Bitcoin Wallet

- It is a digital Wallet. To trade in Bitcoins, a trader must establish a digital wallet. A Bitcoin wallet is analogous to a physical wallet.
- However, instead of storing physical currency, the wallet stores relevant information such as the secure private key used to access Bitcoin addresses and carry out transactions.

Miner

- Generate the blocks
- Confirm the transactions and include in blocks
- Get reward (newly generated bitcoin) and transaction fee from the sender.

Uncle or Stale block

- Block with same number generated by the multiple miners cannot be added in the main blockchain.
- One block will be added in the main chain others will be the uncle blocks.
- Rewards will be provided but no transaction fee.

Bitcoin Blockchain

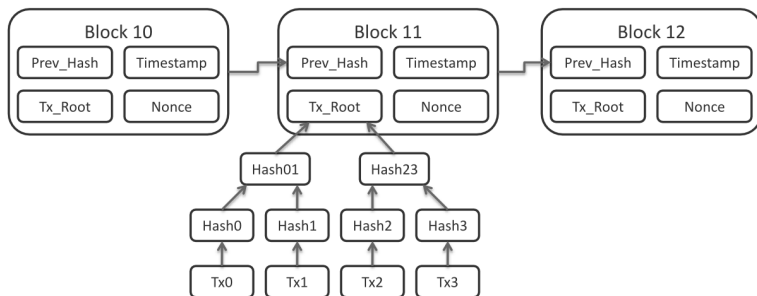


Figure: Blockchain

Bitcoin Header

Version
Previous Block Hash
Merkle Root
Timestamp
Difficulty Target
Nonce

Figure: Header

Blockchain with stale blocks

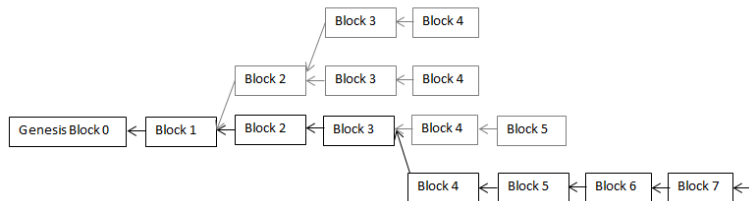


Figure: Blockchain with forks

Way of processing

- UTXO
- Balance

Bitcoin Transaction

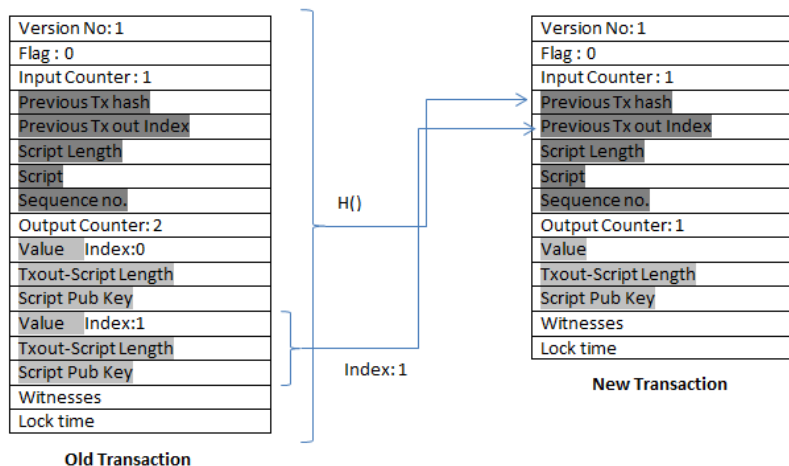


Figure: Bitcoin Transaction overview

Coinbase Transaction

- A transaction, which does not have any parent transaction is referred as coinbase.
- For example, genesis block has *The Times 03/Jan/2009 Chancellor on brink of second bailout for banks* as parent.
- This is for the rewards and will have some arbitrary data as parent.
- It is mandatory for all the blocks to include reward and transaction fee. Also this is the first transaction of every block.

Transaction Age

- Transactions broadcasted for validation and inclusion in the block always points to unspent transaction output (UTXO).
- These block are collected by every full node and put in the transaction pool.
- Miner chooses transactions from the transaction pool based on priority and include in the block after validation. Transactions are prioritized based on the age of UTXO and high value inputs.
- The age of a UTXO is the number of blocks that have elapsed since the UTXO was recorded on the blockchain or number of blocks that are build on top of the UTXO included block in the blockchain.
- Prioritized transactions will be processed and included in the block without any transaction fee if space is available in the block.
- Unprofitable and Dust Transaction

Important Notes

- Difficulty varies at every two weeks that is 2016 blocks
- Maximum is 21 Million BTC
- Header Size - 80 bytes and block maximum size 1 MB
- Confirmation - 7 Blocks
- Transaction Reward
- Reward Halve

Reward Halve



Figure: reward - <https://www.investopedia.com/tech/what-happens-bitcoin-after-21-million-mined/>

Confirmation

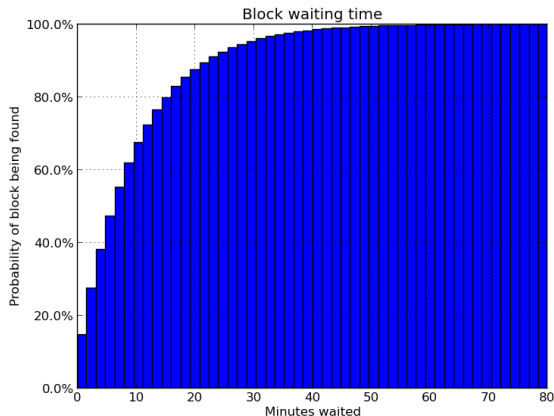


Figure: Confirmation - <https://en.bitcoin.it/wiki/Confirmation>

Proof of Work (PoW)

- This is to control the rate at which blocks can be generated
- This is added in the bitcoin to make the miners to do some work before constructing the blocks.
- Due to the very low probability of successful generation, this makes it unpredictable which worker in the network will be able to generate the next block.

$$n \leq 2^{256}/H_d$$

Proof of Stake (PoS) — Proof of Burn (PoB) [For other coin] — Proof of Elapsed Time (PoE)

- To reduce the energy consumption.

Hash Generation with respect to Proof of Work

Table: Hash Generation

Target ($<$)	<i>Real time (s)</i>	<i>User time (s)</i>	<i>Attempts</i>
2^{210}	> 103146.837	> 397295.636	-
2^{220}	3468.381	13006.832	$\sim 2^{32}$
2^{230}	15.03	57.378	18960126
2^{240}	0.289	0.356	1050648
2^{250}	0.295	0.328	1033

Bitcoin Difficulty

The new difficulty cannot be more than four times of old difficulty and not less than $\frac{1}{4}$ th of old difficulty ($\frac{old_Difficulty}{4} \geq Difficulty \leq old_Difficulty * 4$).

$$Difficulty = \frac{Max_Target}{Current_Target} \quad (1)$$

Max_Target =

```
0X00000000FFFF00000000000000000000000000000000  
0000000000000000
```

$$Deviation = \frac{Expected_Time}{Actual_Time} \quad (2)$$

$$Difficulty = Deviation * old_Difficulty \quad (3)$$

$$Current_Target = \frac{Max_Target}{Difficulty} \quad (4)$$

Sybil Attack

- A node with Multiple Identities.
- It is not possible because of proof of work

Double Spending

- A coin spend more than once.

Eclipse Attack

- In an eclipse attack, an attacker takes control of all the connections going to and from a targeted victim's node. This way, an attacker prevents that victim from obtaining full information about other parts of the network.
- It stores based on the distance.
- If it does not find any node then it asks the neighbor.

Information

- <https://ethstats.net/>
- <https://bitinfocharts.com/>

References

- [1] Satoshi Nakamoto *Bitcoin: A Peer-to-Peer Electronic Cash System* White paper.2008.
- [2] *Transaction* <https://en.bitcoin.it/wiki/Transaction>
- [3] [https://en.wikipedia.org/wiki/Blockchain_\(database\)](https://en.wikipedia.org/wiki/Blockchain_(database))
- [4] <https://github.com/ethereum/wiki/wiki/Mining>
- [5] Karl Wußt and Arthur Gervais, Do you need a Blockchain?,2017.
<https://eprint.iacr.org/2017/375.pdf>

Thank You!